

Daniel Mielnik

Summary of Qualifications

- Creative and analytical thinker with a passion for 3D modelling, demonstrated through a strong 3D design portfolio.
- Experienced in a wide range of technical skills, with a proficiency in multiple CAD softwares and coding languages.
- Driven worker with strong collaboration skills demonstrated through impactful group work and internships.

Education

Bachelor of Engineering - Aerospace Engineering | Minor in C.S.

Toronto Metropolitan University (Ryerson University)

Sep 2021 - Apr 2025

Relevant Courses: Aerodynamics, Aerospace Structural Design, Aircraft Performance, Gas Dynamics, Machine Learning, Propulsion, Space Robotics

Skills

CATIA V5 | SOLIDWORKS | Fusion360 | AutoCAD | MATLAB | Python | Java
Simulink | ANSYS | C | SQL | Automation | Microsoft Office Suite | Teamwork

Work Experience

Industrial Engineering Intern | MHICA May 2024 - Aug 2024

- Reworked and improved **10 shop aids** in **SOLIDWORKS** with measurements taken by hand and **laser scanning**, which were then **3D printed** and used to **decrease assembly line operator errors** caused by the former aids.
- Drove the initiative to independently **develop a script** that scans for errors in MES input, replacing a flow that required individuals to manually inspect **4,000+ entries per aircraft**, and ensuring reliable incident diagnosis data.
- Automated the generation and distribution of daily project status reports using **MS PowerAutomate** and **PowerBI**, then advised team leads how to automate further tasks, ultimately saving the team **over 15 hours per week**.

Software Developer Summer Intern | RBC May 2023 - Aug 2023

- Developed a website leveraging **IBM WatsonX Generative AI** to summarize customer claims and estimate insurance payouts, resulting in a **50% increase** in number of **claims processed** per day and improved customer satisfaction.
- Migrated daily customer report generation to **Microsoft Report Builder (SSRS)** using **SQL databases/queries** to summarize customer banking information and add search by date functionality to **improve report analysis** workflow.

Technical Systems Analyst Intern | RBC May 2022 - Aug 2022

- Used **RedHat Ansible** to automate data cleanup for quarterly banking statements, **reducing the lag time** from statement availability to clients by up to **3 weeks per quarter**.
- **Automated process monitoring** systems using **Bash on Unix** to alert support teams when critical services are down, replacing dependence on customer reports with the ability to automatically **detect system outages within 5 minutes**.

Interests

3D Printing | Skiing | Hiking | Aviation | Space | Tennis | Robotics | Science Fiction | Architecture | Music | Vinyl | CNC | Travel | Drawing | Reading

[Portfolio](#) | 416-779-6410

[LinkedIn](#) | dm.danielmielnik@gmail.com

Academic Achievements

Bombardier Aircraft Design Project ✈️

Jan 2025 - Apr 2025 | 1st Place

- Collaborated with a group of 13 individuals to **design a multi-role aircraft** for the Canadian government with SAR capabilities, including passenger, cargo, and military configurations.
- Led a team of four in aircraft CAD development by coordinating specifications through **CATIA V5** variables and a **structured DMU**, iterating on feedback from Bombardier engineers to model masterlines and structural components.
- Developed a **Matlab** app that enables users to modify the aircraft wing planform area and view its related aerodynamic properties, **accelerating design exploration** and adjustments by removing the need to perform manual computations.

Mixed-Reality Immersive Motion Simulation (MIMS) Lab Research Assistant

Sep 2024 - Dec 2024

- Implemented accurate, up-to-date taxiway routes for takeoff and landing scenarios at CYYZ and CYTZ within an in-development custom flight simulator.
- Mapped the routes provided by ATC using geo-locational data from Google Earth, then transformed the data to a format compatible with **Unreal Engine 5**, where the route was plotted.

MARS Rocket Competition 🚀

Sep 2024 - Dec 2024 | 4th Place

- Working on a team of 5 individuals to iteratively design, build, and test a model rocket; on launch day the rocket reached an **apogee of 440 m** and was successfully **retrieved undamaged**.
- Conducted a thorough **material analysis** for rocket components: chose LW-PLA for the nose cone and transition tube for its light mass, and fiberglass/birch for the fins for structural rigidity and aversion to heat-induced deformations.
- Designed rocket fins, motor mount, and cone, utilizing **OpenRocket** for preliminary design analysis to ensure proper centers of pressure and gravity, then modeled the components in **Fusion360** and **ONSHAPE**.